Alice agreed. “No question. Von Neumann realized that there was a mathematical equivalence between a classical situation when we’re unsure about the exact state of a system, which gives rise to entropy, and the quantum situation where two subsystems are entangled, so we can’t talk about the wave function of either piece separately. He derived a formula for the ‘entanglement entropy’ of a quantum system. The more entangled something is with the rest of the world, the higher its entropy.”

“Aha,” exclaimed her father excitedly. “I see where you’re going with this. The fact that wave functions only branch forward in time and not backward is not simply reminiscent of the fact that entropy increases—it’s the same fact. The low entropy of the early universe corresponds to the idea that there were many unentangled subsystems back then. As they interact with each other and become entangled, we see that as branching of the wave function.”

“Exactly,” Alice responded, with something like daughterly pride. “We’re still not sure why the universe is like that, but once we accept that the early universe was in a relatively unentangled, low-entropy state, everything else follows.”

“But wait a minute.” Her father seemed to have just realized something. “According to Boltzmann, entropy is only likely to increase, it’s not an absolute rule. It’s ultimately due to the random motions of atoms and molecules, so there’s a nonzero probability that entropy will spontaneously go down. Does that mean that it’s possible that decoherence will someday reverse, and worlds actually will fuse together rather than branching apart?”

“Absolutely,” said Alice with a nod. “But just like with entropy, the chance of that happening is so preposterously small that it’s irrelevant to our daily lives, or to any experiment in the history of physics. It’s extremely unlikely that two macroscopically distinct configurations have recohered even once in the lifetime of our universe.”

“So you’re saying there’s a chance?”

“I’m saying that if your worry about Many-Worlds is that branches of the wave function will someday come back together, you’ve clearly exhausted all the reasonable worries and are grasping at straws.”